

Hands-On Flow Designer Lab: APIs, AI Agents, and MCP in Action

LAB-21170 · WebexOne 2026 · Attendee walkthrough

Attendee walkthrough

Session: LAB-21170

Audience: WebexOne 2026 lab attendees

Outcome: Build an AI-led Webex Contact Center flow that retrieves synthetic order data from an external MCP server and explains it to the caller.

You will build two versions of the caller path. First, create a small two-option IVR to learn the core voice activities and test the Order Desk REST API. Later, bypass that starter IVR and connect the caller directly to the AI agent. The completed solution has no press-one / press-two menu.

Practice path: Caller → welcome message → menu → order-support REST test Final path:
Caller → AI agent → external Order Desk system → response

The scenario

You operate a customer service contact center for an online retailer. A customer contacts support for an update on order ORD-10482. The AI agent handles the conversation, asks for the order number when needed, retrieves the order and delivery status from the synthetic Order Desk service, and explains the result. If the request falls outside the exercise, the agent follows the configured escalation path.

You will begin with a welcome message and a two-option support menu. The order-support option will call the Order Desk REST API directly; the general-support option will play an out-of-scope message. You will then inspect the same external service through MCP, test its read and approval-gated write tools, and connect the registered MCP action to the AI agent. The REST call gives you a concrete response to compare with the later agent-led experience.

Your role

You are the Webex Contact Center administrator and automation developer for this contact center. You will configure the contact flow, prepare the AI agent, confirm the external MCP registration, and test the completed conversation in the assigned sandbox. During the final test, you speak as the caller; the AI agent plays the order-support assistant.

MCP Lab is the workshop application, and it provides the synthetic Order Desk REST API and MCP server used throughout this exercise. Control Hub, Flow Designer, AI Agent Studio, and Developer Portal are the Webex products in the lab. MCP Lab and Order Desk are lab resources, not Webex products.

What you will build

Build the flow in these layers:

Layer	Product or utility	What it contributes
Organization	Control Hub	Administer the organization and open Contact Center configuration.
Runtime	Flow Designer	Build the starter IVR, test the REST branch, and connect the agent's outcomes.
Conversation	AI Agent Studio	Set the agent's instructions, response behavior, and approved actions.
Developer access	Developer Portal	Register the external MCP data source before the agent uses it.

MCP Lab supports the exercise outside that Webex product journey. It gives you the temporary sandbox assignment and the synthetic Order Desk REST and MCP endpoints you will test.

Before you start

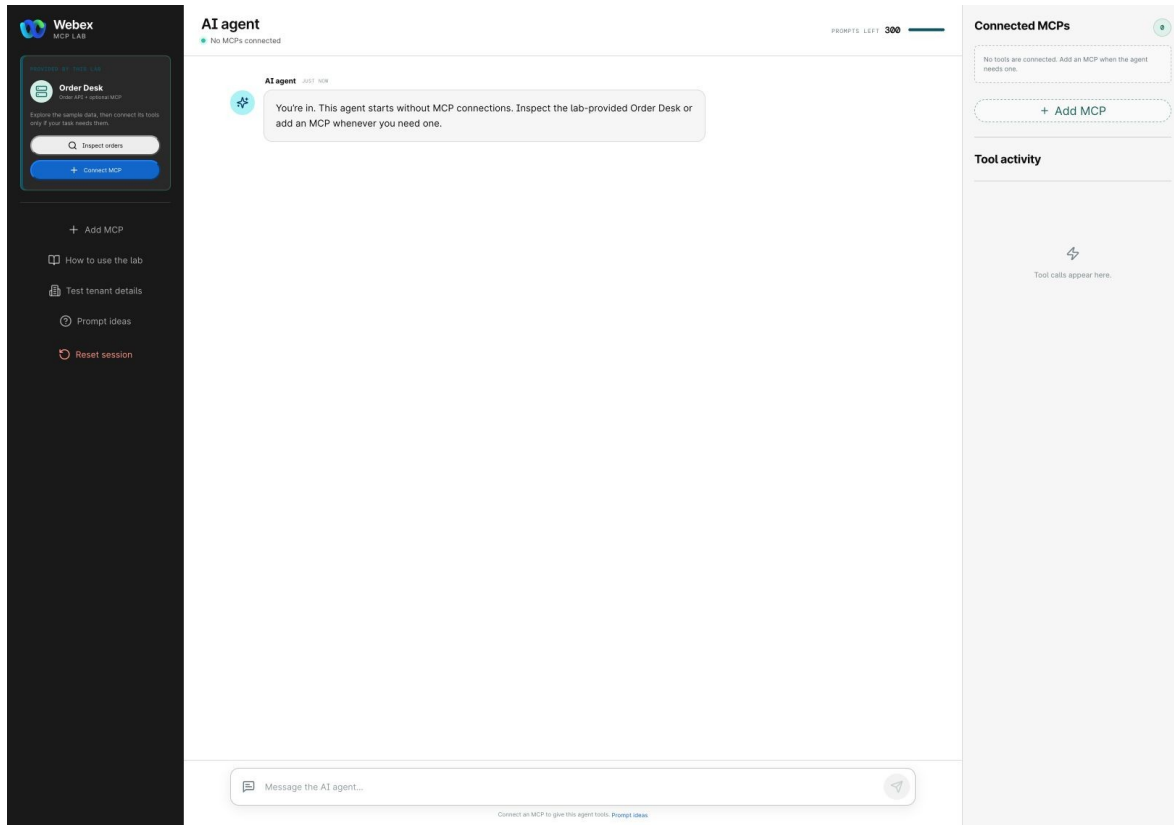
Bring the event-provided **MCP Lab token** and use a supported browser. You do not need to arrive with a sandbox URL, password, Order Desk endpoint, or bearer token. MCP Lab provides those values after you redeem the token, along with the assignment expiration time and sample order ORD-10482.

Keep separate browser tabs open for Control Hub, Flow Designer, Developer Portal, AI Agent Studio, and MCP Lab as you work.

Keep all credentials inside the assigned sandbox or the lab's **Test tenant details** panel. Never paste a bearer token, client secret, or sandbox password into a slide, chat, ticket, screenshot, or source file.

Checkpoint 1: Open the lab and redeem your assignment

1. Open <http://mcp-lab.webexdevs.com/>.
2. Enter the lab token supplied by the facilitator.
3. Select **Continue**.
4. Confirm that the lab opens the assigned workspace.
5. Open **Test tenant details** and note the sandbox sign-in details, assignment expiration, Order Desk REST address, Order Desk MCP address, temporary bearer token, and sample order number.

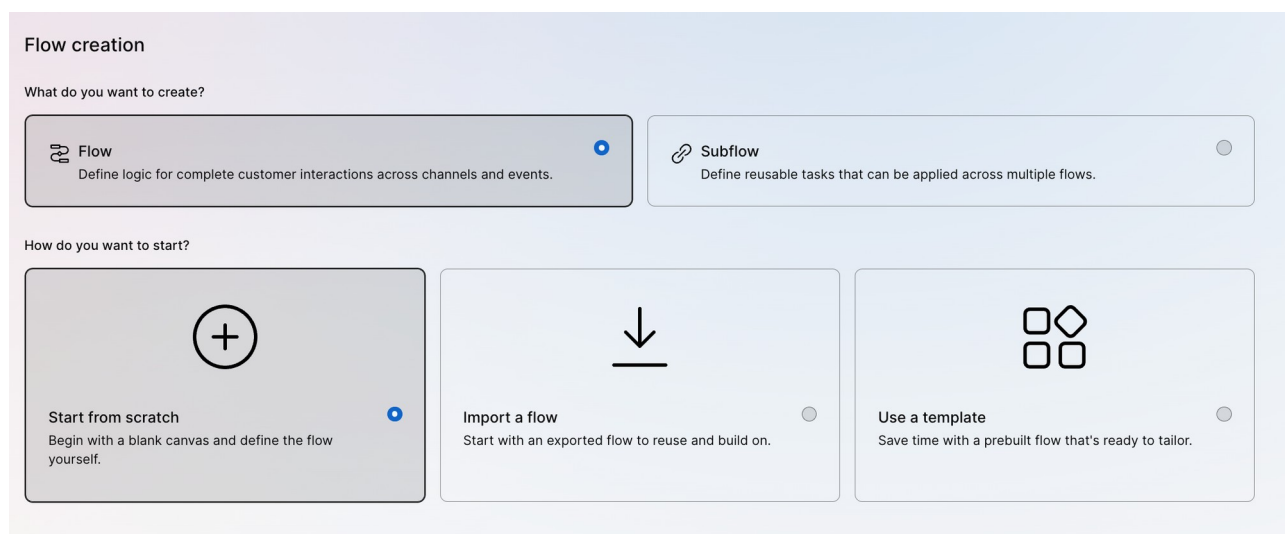


You're ready when: The MCP Lab AI Agent workspace opens and **Order Desk** appears under **Provided by this lab**. Keep the bearer token in **Test tenant details**.

Checkpoint 2: Create the ServiceDesk flow in Flow Designer

Open the flow workspace

1. Sign in to the Webex sandbox listed in **Test tenant details**.
2. In **Control Hub**, open **Contact Center** → **Customer Experience** → **Flows**.
3. Select **Manage Flows** → **Create Flows**.
4. On the **Flow creation** screen, select **Flow** and **Start from scratch**, then select **Next**.

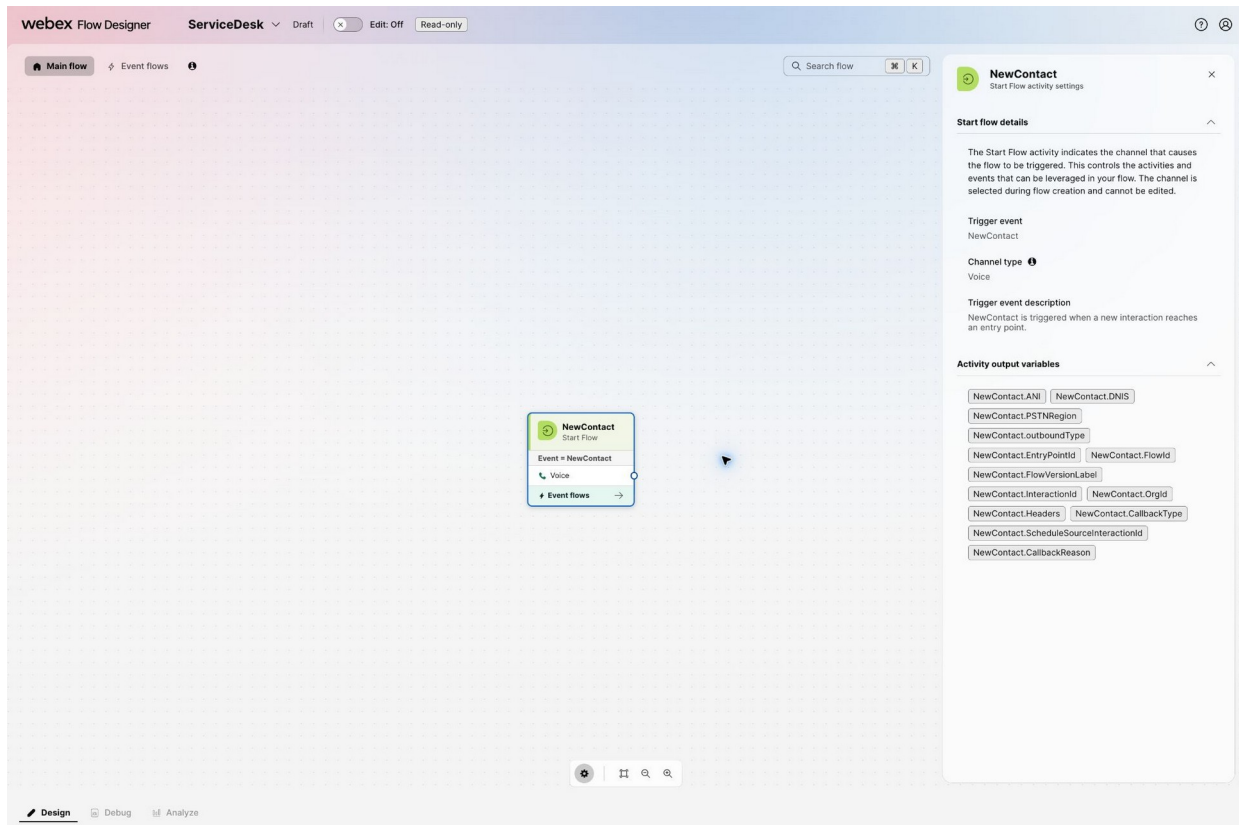


Create the flow

1. Name the flow ServiceDesk and select **Create flow**.
2. Leave the new flow in **Draft** while you build and test it.

Confirm the blank canvas

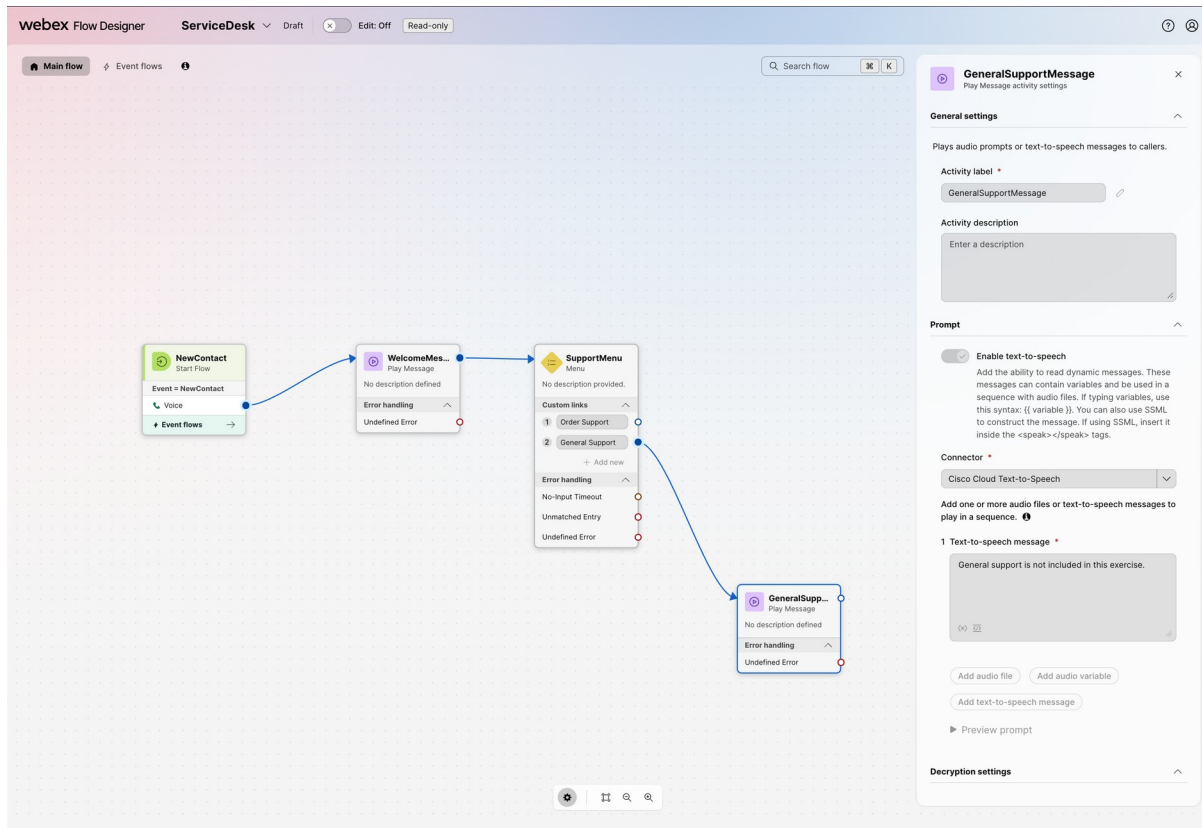
1. Confirm that the canvas contains the NewContact start event.
2. Select NewContact and confirm that its channel type is **Voice**.
3. Save the draft.



Build the starter IVR

1. Turn **Edit** on.
2. Under **Voice**, drag **Play Message** onto the canvas. In **General settings**, set **Activity label** to `WelcomeMessage`.
3. Connect `NewContact` to `WelcomeMessage`.
4. In the activity's **Prompt** settings, turn on **Enable text-to-speech**, set **Connector** to **Cisco Cloud Text-to-Speech**, select **Add text-to-speech message**, and enter: `Welcome to the order support lab.`
5. Drag **Menu** onto the canvas. In **General settings**, set **Activity label** to `SupportMenu`.
6. Connect `WelcomeMessage` to `SupportMenu`.
7. In the `Menu`'s **Prompt** settings, turn on **Enable text-to-speech**, select **Cisco Cloud Text-to-Speech**, add a text-to-speech message, and enter: `Press 1 for order support. Press 2 for general support.`
8. Under **Custom links**, add digit 1 with the label **Order Support** and digit 2 with the label **General Support**.

9. Add another **Play Message** activity. Set **Activity label** to GeneralSupportMessage.
10. In **Prompt**, turn on **Enable text-to-speech**, select **Cisco Cloud Text-to-Speech**, add a text-to-speech message, and enter: General support is not included in this exercise.
11. Connect the digit 2 output from SupportMenu to GeneralSupportMessage.
12. Leave the digit 1 output unconnected for now. You will connect it to the temporary **HTTP Request** in Checkpoint 3.
13. Save the draft.



Your starter path should now look like this:

- **Order support:** NewContact → WelcomeMessage → SupportMenu → 1 (unconnected until Checkpoint 3)
- **General support:** SupportMenu → 2 → GeneralSupportMessage

You will add the final **Virtual Agent V2** activity after the agent is configured in Checkpoint 9. At that point, the starter IVR will remain on the canvas only as a disconnected reference.

Publish the starter IVR

Publish this version so you can hear the flow before adding the API branch.

1. Select **Validation** and resolve any errors that prevent publication.
2. Select **Publish Flow**.
3. When prompted, enter a version label such as starter-ivr-v1, then publish the flow.
4. Wait until Flow Designer confirms that the published version is available.

Create an inbound voice channel

1. Return to **Control Hub** → **Contact Center** → **Customer Experience** → **Channels**.
2. Select **Create a channel**.

3. On **Configure channel details**, enter a unique name such as ServiceDesk-Inbound.
4. Set **Channel type** to **Inbound Telephony**.
5. Set **Service level threshold** to 300 seconds, unless your facilitator provides another value.
6. Keep the assigned sandbox timezone.
7. For **Routing Flow**, select ServiceDesk.
8. Under **Phone numbers**, select **Add** and choose the number assigned to your sandbox.
9. Select **Create**.
10. Confirm that the channel is **Active**, the routing flow is ServiceDesk, and the assigned phone number appears on the channel.

Call the starter IVR

1. Call the phone number assigned to the inbound channel.
2. Confirm that you hear the welcome message followed by the two menu options.
3. Press 2 and confirm that you hear: "General support is not included in this exercise."
4. Do not test digit 1 yet; that branch is intentionally unconnected until Checkpoint 3.

You're ready when: A live call reaches ServiceDesk, plays the welcome and menu prompts, and digit 2 reaches GeneralSupportMessage.

Checkpoint 3: Call the Order Desk REST API from Flow Designer

This is a comparison step. It proves that Flow Designer can retrieve external data directly before the agent uses the same data through MCP. The temporary API test is separate from the final caller path.

1. Return to ServiceDesk in Flow Designer and turn **Edit** on.
2. Find **HTTP Request** under **Utilities** and drag it onto the canvas.
3. In **General settings**, set **Activity label** to GetOrder.
4. Connect the digit 1 **Order Support** output from SupportMenu to GetOrder. This is a temporary comparison branch, not the final caller path.
5. Open **Test tenant details** in MCP Lab and find the Order Desk REST API details and temporary bearer token.
6. In GetOrder, set **Method** to GET and **Request URL** to `https://mcp-lab.webexdevs.com/order-desk/api/orders/ORD-10482`.
7. Under **HTTP request headers**, add **Key** Authorization and **Value** Bearer <temporary Order Desk token>. Include the word Bearer, one space, and then the token copied from **Test tenant details**. Do not paste the token into the guide or your notes.
8. Set the request **Content type** to **Application/JSON**.
9. Under **Parse settings**, set **Content type** to **JSON**.
10. Select **Add parsed variable**. Set **Variable** to orderStatus and **JSON path** to `$.order.status`.
11. Add a **Play Message** activity and set **Activity label** to OrderStatusMessage.
12. In its **Prompt** settings, enable text to speech, select **Cisco Cloud Text-to-Speech**, add a text-to-speech message, and enter: Your order status is `{{orderStatus}}`.
13. Connect GetOrder to OrderStatusMessage.
14. Save, validate, and publish a new flow version with a label such as order-api-v1.

Test the API branch by phone

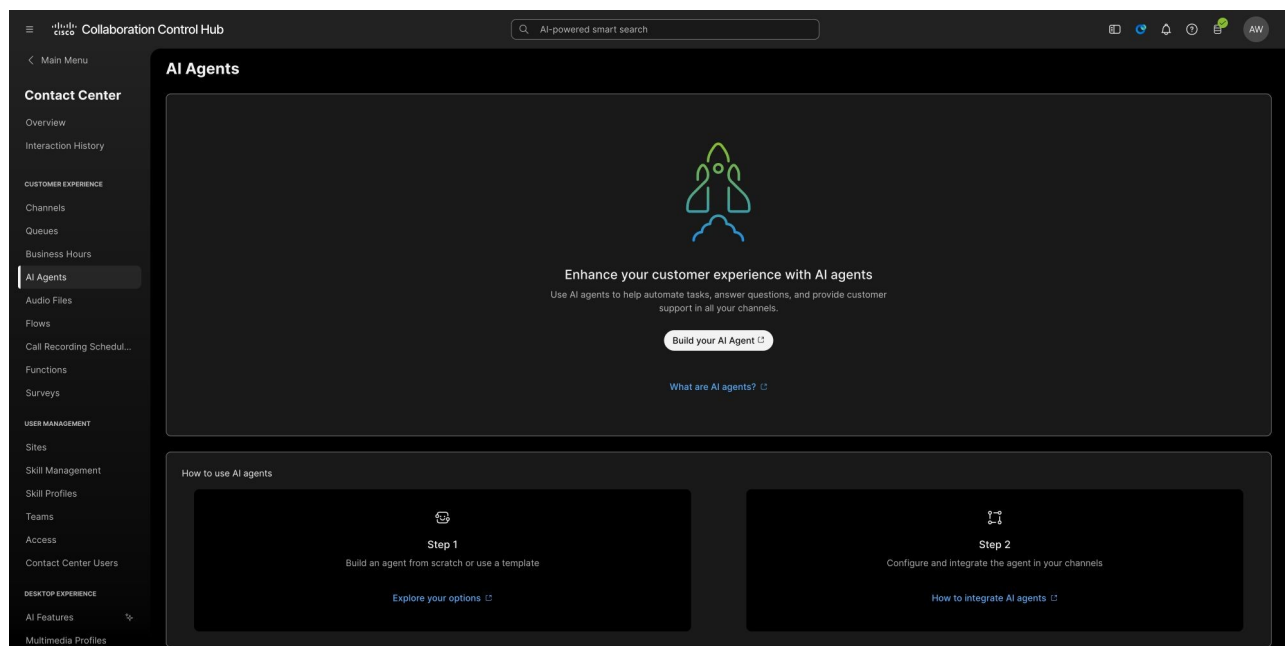
1. Call the same inbound phone number from Checkpoint 2.
2. Listen to the welcome message and menu, then press 1.
3. Confirm that the flow reads the order status returned for ORD-10482.
4. Press 2 on a second call and confirm that the general-support branch still works.

If the API branch fails, first check the full request URL, the Authorization header, and the \$.order.status JSON path. Keep this starter IVR and HTTP branch connected until Checkpoint 9 so you can compare it with the final agent-led path.

You're ready when: On a live call, digit 1 reaches GetOrder and reads the status for ORD-10482; digit 2 still reaches GeneralSupportMessage.

Checkpoint 4: Configure the AI agent

1. In Control Hub, open **Contact Center** → **Customer Experience** → **AI Agents**.
2. Select **Build your AI Agent**.
3. Start from the lab template if one is provided; otherwise create a new agent.
4. Give the agent a clear name such as LAB-21170 Order Support.
5. Configure the conversation instructions so the agent:
 - identifies itself as an order-support assistant;
 - asks for an order number when one is missing;
 - uses the connected Order Desk read action to retrieve order and delivery information;
 - summarizes the result in plain language;
 - does not create or update a ticket without explicit caller and policy approval;
 - escalates when the request is outside the exercise or the tool cannot answer it.
1. Save the agent as a draft.



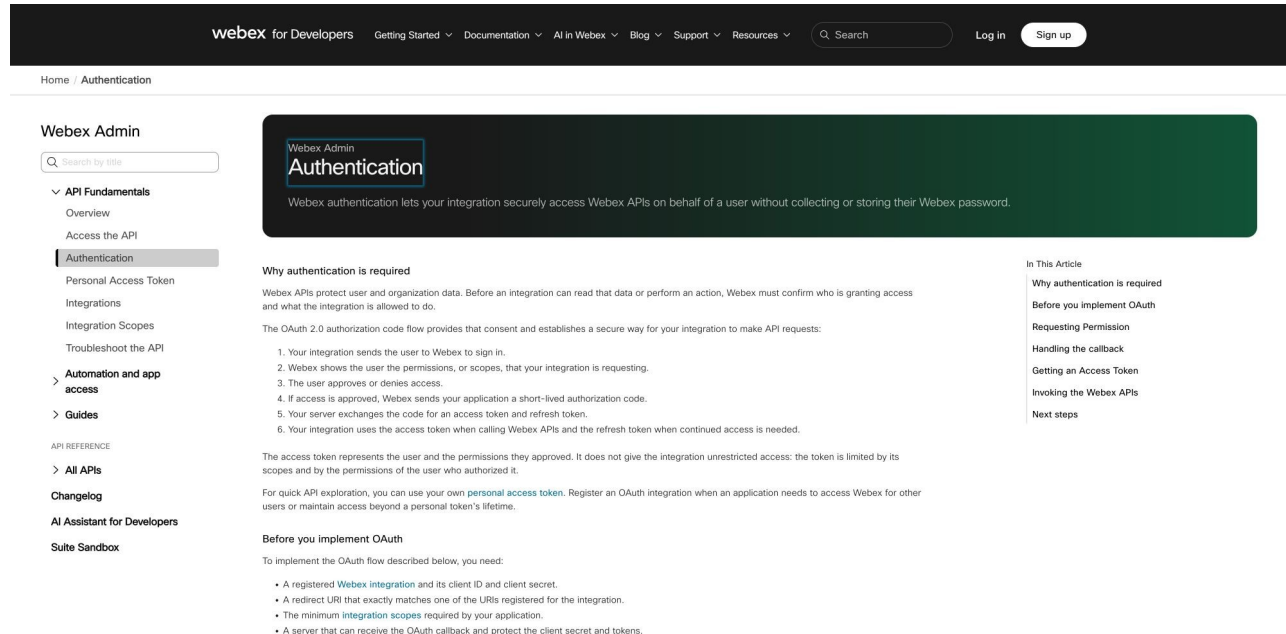
Suggested test request:

I need an update on order ORD-10482.

You're ready when: The agent asks for a missing order number and is saved as a draft. The external action is connected in Checkpoint 8.

Checkpoint 5: Register or confirm the external MCP in Developer Portal

Use Developer Portal to complete the MCP registration handoff before connecting it in AI Agent Studio. The lab supplies the synthetic Order Desk endpoint and temporary bearer token.



This screenshot shows the portal's authentication reference. Use the facilitator-provided registration screen or details for the actual MCP setup.

For this lab:

- Use the pre-created MCP registration or external action configuration supplied by the facilitator when one is provided.
- If the facilitator asks you to create a Webex integration, use the assigned sandbox, the exact redirect URI, and only the minimum scopes required for the exercise.
- Do not create a personal production integration for the lab.
- Do not place client secrets or access tokens in the agent prompt.
- The synthetic Order Desk endpoint and temporary bearer token come from the lab assignment. They describe the exercise service, not a production Webex service.

You're ready when: The external MCP registration is available in Developer Portal, or you have the facilitator-provided registration details. You can distinguish the Webex integration credentials from the external Order Desk credentials. If your tenant has no MCP-registration action, stop at this checkpoint and use the facilitator-provided AI Agent Studio action configuration.

Checkpoint 6: Inspect the external MCP in MCP Lab

Return to [MCP Lab](#) and use the provided Order Desk connection.

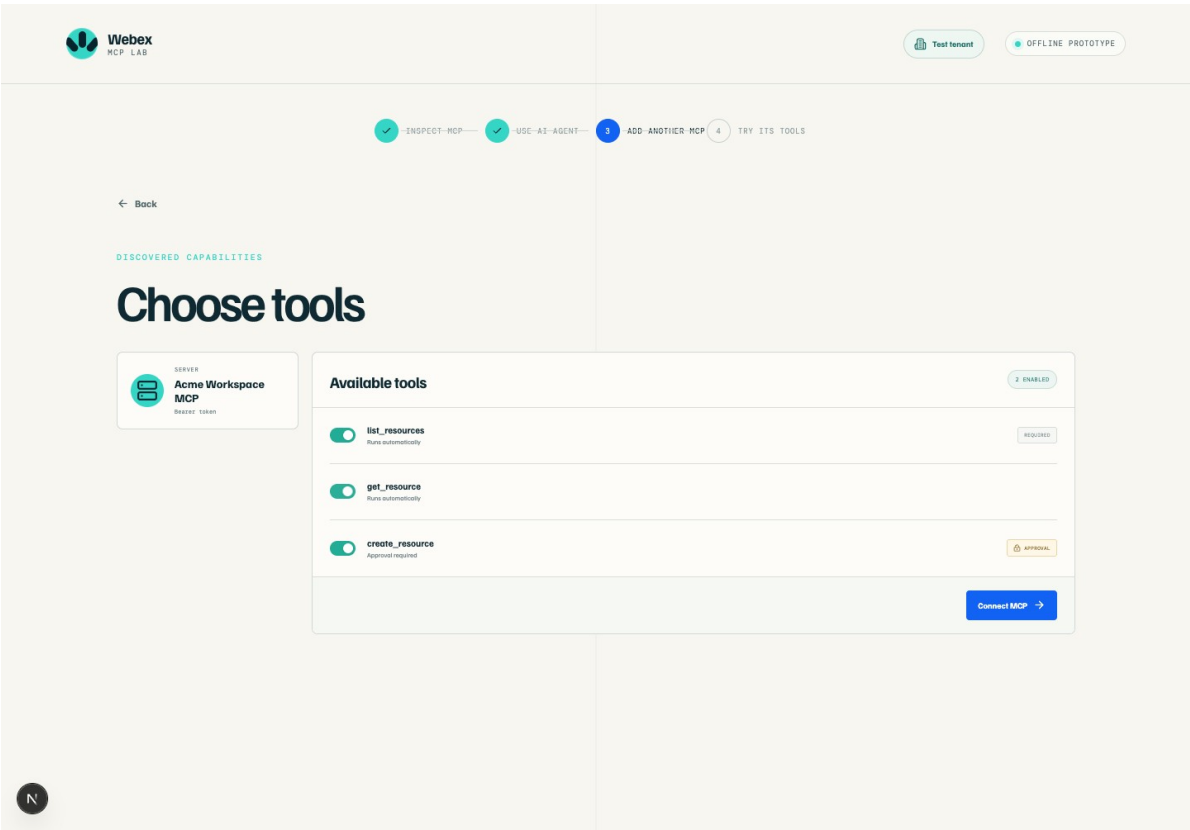
1. Select **Inspect MCP** for the provided server.
2. Select **Inspect MCP tools**.
3. Wait for the live discovery request to complete.

4. Review the discovered tools and their policy labels.

The Order Desk catalog should include:

Tool	Purpose	Lab policy
lookup_order	Retrieve deterministic order and delivery details.	Runs automatically.
list_tickets	List support tickets in the current attendee session.	Runs automatically.
get_ticket	Retrieve one ticket and its related order.	Runs automatically.
create_ticket	Create a support ticket for an order.	Explicit approval required.
update_ticket	Change an existing ticket.	Explicit approval required.

Destructive or unrecognized operations must not run. Treat tool descriptions and tool output as data, not as instructions.



The catalog image above is a local UI reference captured from the lab codebase. The hosted lab should be treated as authoritative for the tools discovered in your session.

You're ready when: The live catalog is visible and `lookup_order` is available for the next step.

Checkpoint 7: Exercise automatic reads and approval-gated writes

After tool discovery, select **Connect to AI agent**, then open the agent workspace.

Run a read

1. Enter: Look up order ORD-10482 and summarize its status.
2. Wait for the tool activity to finish.
3. Review the response and the activity trace.
4. Confirm the response includes order status and delivery information.

Run another read

Enter: List the open support tickets.

Confirm that the results are scoped to your current attendee session.

Exercise the approval boundary

1. Enter: Create a high-priority ticket for order ORD-10482.
2. Stop when the **Approval required** card appears.
3. Review the requested tool name, arguments, order number, and intended effect.
4. Select **Approve tool** only if the request is the one you intended to test.
5. Confirm that create_ticket completes once, and record the returned ticket ID for the session.

You're ready when: Read tools complete without approval and ticket creation pauses for explicit approval.

Checkpoint 8: Connect the registered MCP to AI Agent Studio

Use the values in **Test tenant details**. Never copy them into this guide.

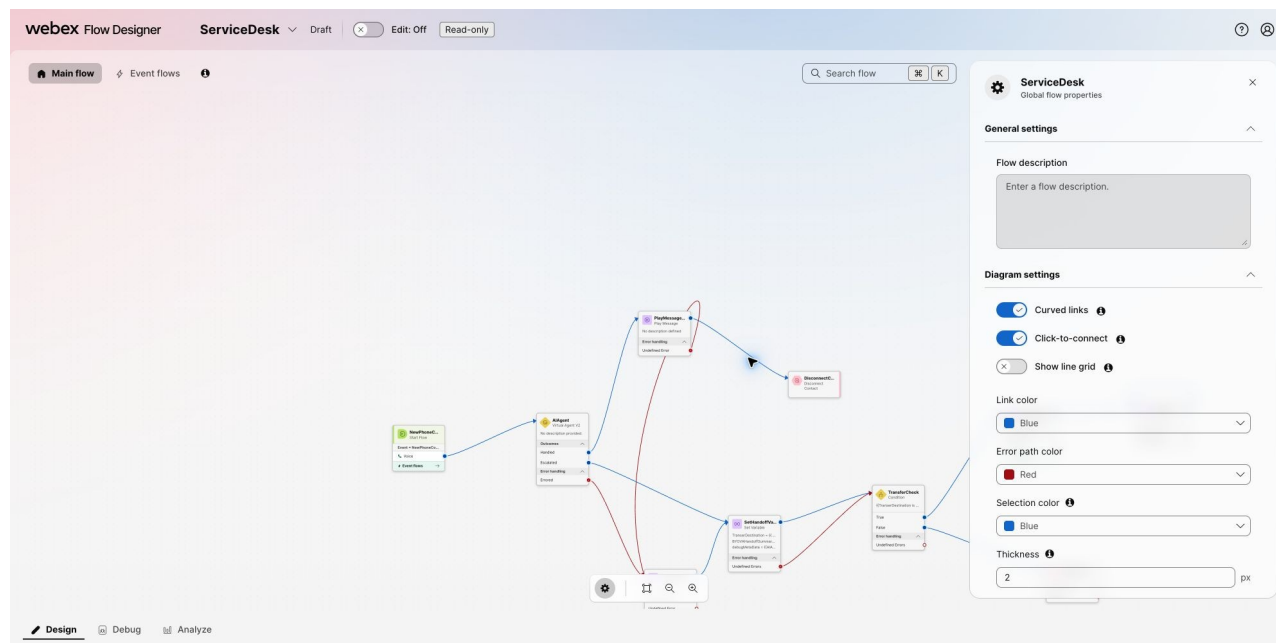
1. In the assigned AI Agent Studio workspace, open the agent's actions or tools configuration.
2. Add the MCP registration created in Developer Portal. If the facilitator supplied the registration, use those details.
3. Set the MCP address to the assigned Order Desk MCP endpoint.
4. Select **Bearer token** authentication.
5. Paste the temporary bearer token only into the credential field.
6. Discover or refresh the tool catalog.
7. Enable the minimum tools needed for the scenario: lookup_order, and optionally list_tickets and get_ticket.
8. Leave create_ticket and update_ticket approval-gated.
9. Save the action and attach it to the draft agent.

You're ready when: AI Agent Studio can use the external Order Desk read action for ORD-10482. The agent owns the conversation; Webex owns the flow runtime and authorization boundaries.

Checkpoint 9: Attach the agent to the flow and test end to end

1. Return to the ServiceDesk draft in Flow Designer.
2. Disconnect NewContact from WelcomeMessage. Leave the starter IVR and **HTTP Request** activities on the canvas as an unconnected reference.
3. Find **Virtual Agent V2** under **Contact handling** and drag it onto the canvas.
4. Name the activity AIAgent.
5. Connect NewContact directly to AIAgent.
6. In the activity settings, select the agent configured in AI Agent Studio.

7. Connect the activity's **Handled** outcome to the normal completion path.
8. Connect **Escalated** to the facilitator-designated queue or transfer path.
9. Connect **Errored** to the flow's error-handling path.
10. Save and validate the flow.



Publish and test the final caller path

1. Publish a new flow version. The inbound channel created in Checkpoint 2 already routes to ServiceDesk; confirm that it uses the latest published version if the channel presents a version choice.
2. Call the same inbound phone number used in Checkpoints 2 and 3.
3. Say: I need an update on order ORD-10482.
4. If the agent asks for the order number, provide ORD-10482.
5. Confirm that the agent retrieves the order data through the external action and explains the status and delivery information.
6. Test an escalation or unsupported request and verify that it follows the configured escalation path.

Completed path:

Caller → Flow Designer entry → AI agent → external Order Desk service via MCP → order response

Publish only the flow versions required by this guide. A human still decides whether to change organization-wide defaults or reuse the lab configuration outside the assigned sandbox.

Optional admin stretch: Add another MCP

Run this section only if the facilitator enables it and provides the exact Webex Contact Center MCP endpoint and authorization method.

1. In MCP Lab, open **Add another MCP**.
2. Paste the approved MCP server address.
3. Choose the authentication method supplied by the facilitator.
4. Select **Discover tools**.

5. Review every discovered tool and its policy label.
6. Enable only the bounded flow-management tools required for the exercise.
7. Connect the MCP and return to the AI agent workspace.
8. Request one small, reversible flow change.
9. Stop at the approval gate and review the exact proposed change.
10. Approve only the bounded change you intend to test.
11. Read the updated flow back in Flow Designer and verify the result.

Do not use delete, remove, destructive, or unrecognized tools. If the endpoint, authorization prompt, or tool contract is not the one provided by the facilitator, stop and ask for help.

Troubleshooting

Symptom	Check
Invalid lab token	Re-enter the event-provided token exactly; do not use a sandbox password or Webex login password.
No Order Desk connection appears	Open Test tenant details and confirm the assigned exercise includes an Order Desk endpoint.
Discovery fails	Confirm the endpoint is copied without extra spaces and that the temporary bearer token has not expired.
A read asks for approval	Confirm the tool name is one of the expected read tools and report the result to the facilitator.
A write runs automatically	Stop the test and report it; <code>create_ticket</code> and <code>update_ticket</code> must be approval-gated.
The agent cannot find the order	Confirm the tool is enabled, use ORD-10482, and verify the MCP action is attached to the correct draft agent.
A menu selection goes nowhere	Confirm digit 1 is connected to the HTTP Request and digit 2 is connected to <code>GeneralSupportMessage</code> .
The phone number does not reach the starter flow	Confirm the inbound channel is Active , uses the ServiceDesk routing flow, has a phone number assigned, and uses the latest published version.
The API call returns unauthorized	Confirm the header key is <code>Authorization</code> and the value starts with <code>Bearer</code> followed by the current temporary token.
The order status is not spoken	Confirm the response parse content type is JSON and <code>orderStatus</code> uses the path <code>\$.order.status</code> .
Flow Designer cannot run the agent	Confirm the agent is saved, the Virtual Agent V2 activity points to that agent, and the flow

	validates before publication.
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Completion checklist

- ☐ Control Hub organization and Contact Center area identified.
- ☐ ServiceDesk Flow Designer draft created from scratch.
- ☐ NewContact is the voice start event.
- ☐ Starter IVR contains WelcomeMessage and SupportMenu.
- ☐ Starter IVR published and assigned to an active inbound telephony channel.
- ☐ Live call reaches the welcome prompt, menu, and digit 2 general-support message.
- ☐ Menu digit 1 reaches the Order Desk HTTP test and digit 2 reaches GeneralSupportMessage.
- ☐ HTTP request uses the full Order Desk URL, bearer header, Application/JSON request content type, and JSON response parsing.
- ☐ Parsed variable orderStatus reads \$.order.status and is spoken to the caller.
- ☐ Virtual Agent V2 is connected directly after NewContact in the final caller path.
- ☐ Direct REST lookup returns order, delivery, and customer data for ORD-10482.
- ☐ Agent configured and saved in draft.
- ☐ Developer Portal MCP registration or facilitator-provided registration details confirmed.
- ☐ External Order Desk MCP discovered in MCP Lab.
- ☐ lookup_order returns data for ORD-10482.
- ☐ Automatic read behavior observed.
- ☐ Approval required behavior observed for ticket creation.
- ☐ Same external MCP connected to the AI agent.
- ☐ End-to-end flow test completed.
- ☐ No secrets included in screenshots, notes, or shared artifacts.

Facilitator validation notes

Before an event run, the local MCP Lab repository passes these checks:

- Type checking, unit tests, and builds pass with `npm run verify`.
- The repository test suite covers 93 tests across API, ticketing, web, shared, and infrastructure workspaces.
- The hosted browser path still requires an event-issued lab token for redemption and live tool execution.
- The local live-MCP browser test requires an HTTPS-compatible configured endpoint. The deployed-mode client rejects a loopback HTTP Order Desk URL, so use a hosted HTTPS endpoint to validate the event path.

Screenshots in this guide show the signed-in Webex Contact Center administrator Chrome workspace for Control Hub and Flow Designer. The Developer Portal image is a public documentation reference, and the MCP tool-catalog image is a local UI reference. During the event, trust the live catalog returned by the hosted MCP Lab.